Atom Display

SKU:K115













Description

Atom Display is a low-cost display driver that uses FPGA to simulate traditional SPI TFT-LCD data output, supporting a maximum image output of 720P (1280 x 720). It incorporates a dedicated RGB-to-HD multimedia signal conversion chip for wide compatibility in display output. Embedded with an ESP32 controller, it integrates 2.4G Wi-Fi and comes with 8MB Flash + 2MB PSRAM memory. **Atom Display** can replace traditional PC-controlled display panel solutions, offering a more cost-effective alternative.

Features

- FPGA and driver library code are open-source
- Utilizes FPGA (Gowin GW1NR-9C)
- Built-in LT8618SX RGB-to-HD multimedia signal conversion chip (supports 24-bit color depth)
- SPI interface (FPGA) + I2C interface (LT8618SX)
- Maximum image output of 720P (1280x720)
- Multiple output modes, optimized frame rate up to 12 ~ 16 FPS
- Built-in Atom PSRAM controller (ESP32-PICO-V3-02, 8MB Flash + 2MB PSRAM)
- Programmable RGB LED x1, Reset button x1, Button x1, Full-featured Grove expansion interface x1
- Development platforms: Arduino, ESP32-IDF (UiFlow not yet supported)

Includes

- 1 x Atom-PSRAM
- 1 x Atomic Display Base

Applications

- Display input signal source
- High-definition data dashboard

Specifications

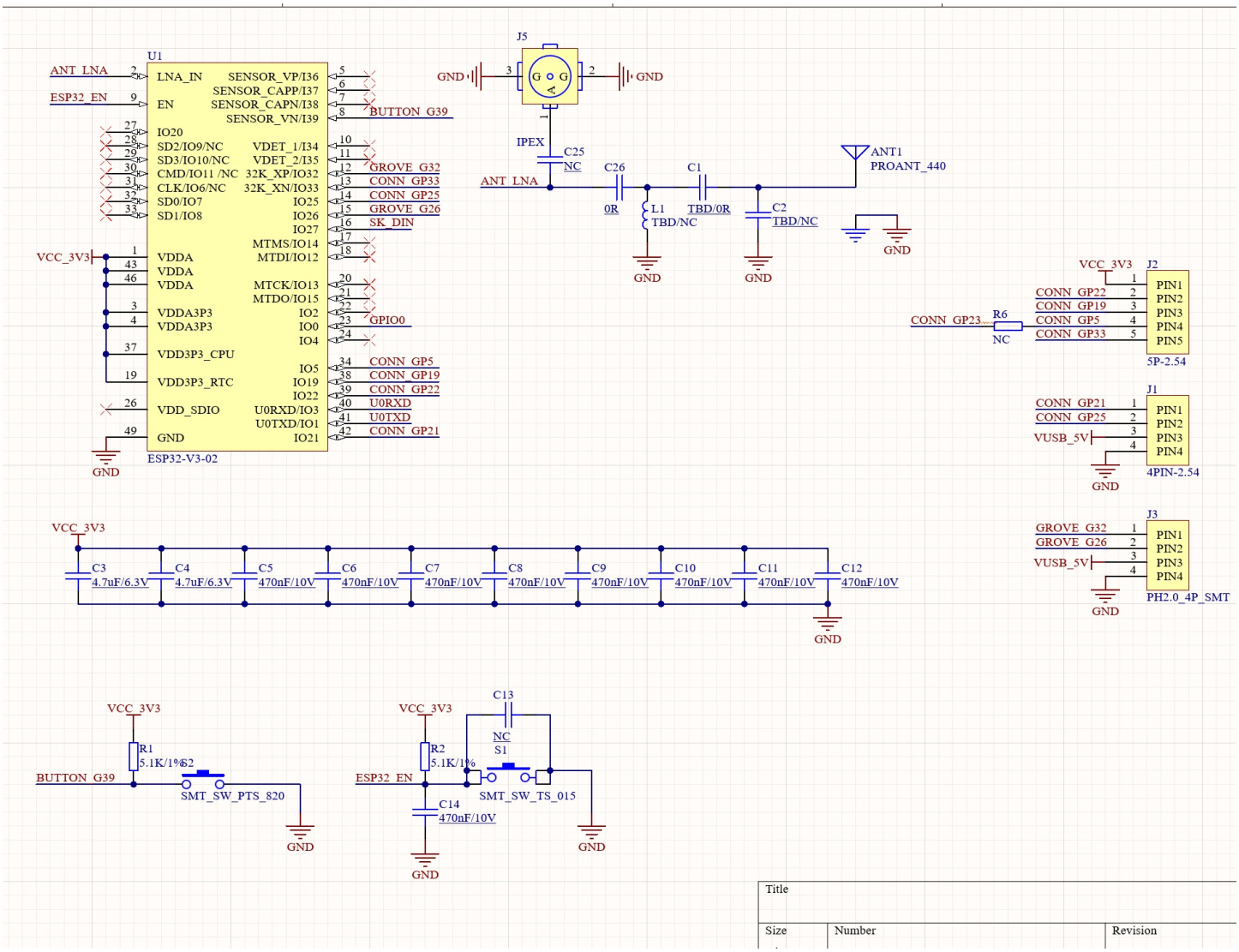
Specification	Parameter
FPGA	Gowin GW1NR-9C
LT8618SX	RGB-to-HD multimedia signal chip, supports 24-bit color
Maximum image output size	720P (1280x720)
Output frame rate	1280x720 60Hz
USB driver chip	CH9102
Product Size	64 x 24 x 29mm
Product weight	21.0g
Package Size	76 x 46 x 29mm
Gross Weight	34.0g

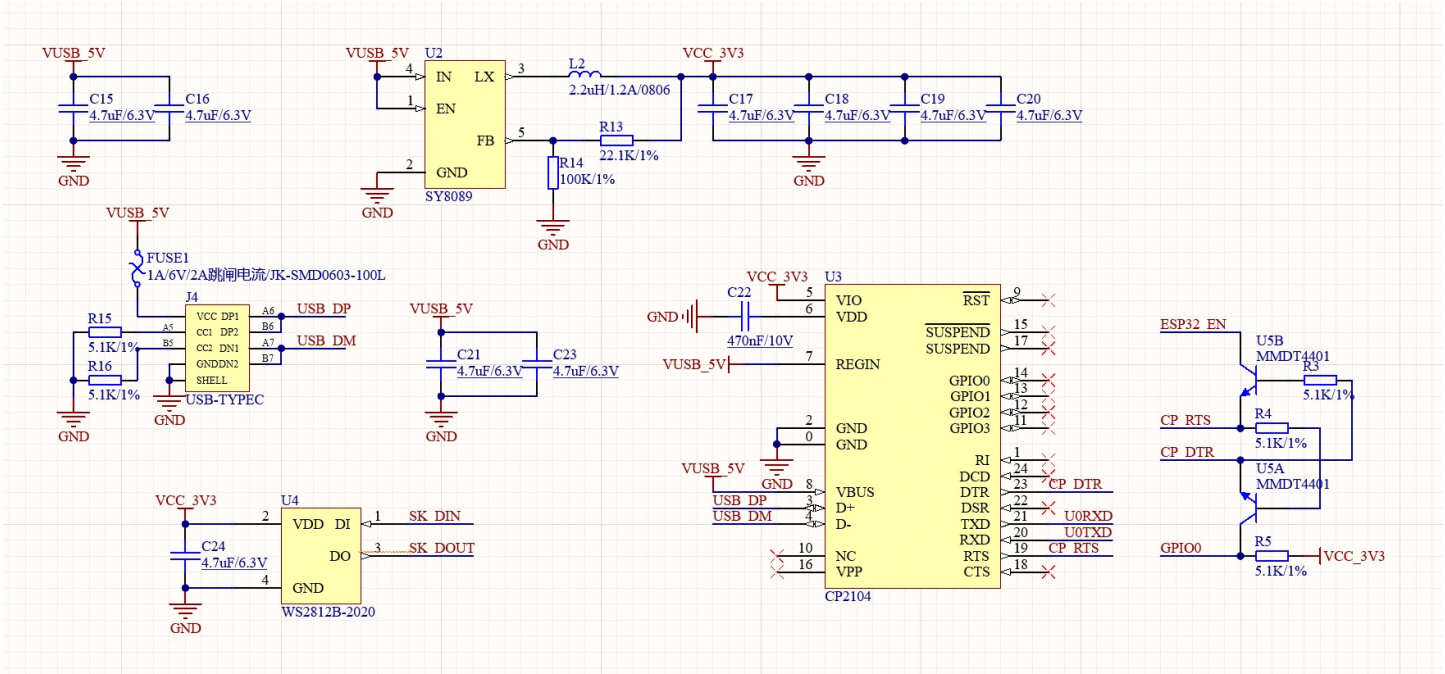
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Display Compatibility

Atom Display Lite requires a display with adaptive resolution scaling functionality. On some displays that do not support adaptive resolution, display anomalies may occur.

Schematics





- Complete Atom Display Schematic

PinMap

PIN	LEFT	RIGHT	PIN
3V3		1	
SPI_CLK	2	3	
CS	4	5	5V
SPI_MISO	6	7	GND
SPI_MOSI	8	9	

- BUTTON & RGB LED

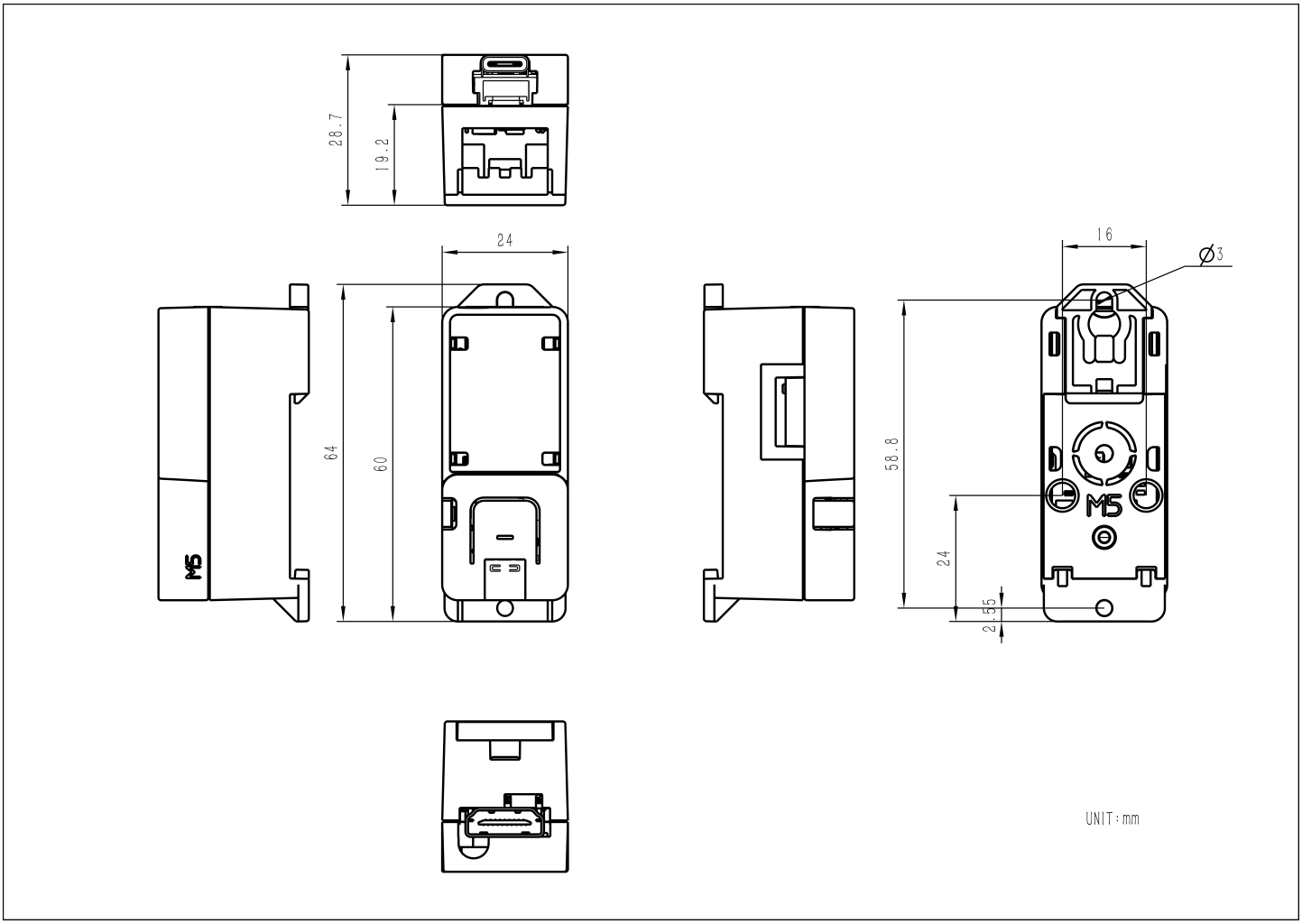
Atom	G39	G27
BUTTON	SIGNAL	/
RGB LED	/	SK_DIN

- LT8618SX

Atom	G25	G21	5V	GND
LT8618SX	LT_CSDA	LT_CSCL	VIN	GND

Model Size

- Atom Display Model Size PDF



Softwares

Arduino

- M5GFX - Lib

Programming Notes

Before using the example programs in the M5GFX library, you need to change the included header file to match the device you are using. As shown in the example below, include `#include <M5AtomDisplay.h>` and create an instance `M5AtomDisplay display;`

```
#include <Arduino.h>
#include <vector>

#include <M5AtomDisplay.h>
M5AtomDisplay display;

void setup(void)
{
  display.begin();
}

void loop(void)
```

```
{
  display.fillScreen(RED);
  delay(1000);
  display.fillScreen(GREEN);
  delay(1000);
  display.fillScreen(BLUE);
  delay(1000);
}
```

Video

ATOM_DISPLAY_VIDEO.mp4